CSE4088 – Introduction to Machine Learning

Homework 4

SVM with Soft Margins

2.

```
Question 2-3
0 0.10588396653408312
1 0.0144013166918118 22
2 0.10026059525442327
3 0.09024825126868742
4 0.08942531888629818
5 0.07625840076807022
6 0.09107118365107666
7 0.08846523110684405
8 0.07433822520916199
9 0.08832807570977919
```

Figure 1 n vs all in-sample error rates

According to the results above **[a] 0 versus all** has the highest Ein among when n in 0, 2, 4, 6, 8.

3.

According to the results above **[a] 1 versus all** has the highest Ein among when n in 1, 3, 5, 7, 9.

4.

```
Question 4
Difference between number of support vectors for 0 and 1: 1793
```

Figure 2 # of support vectors difference for 0-vs-all and 1-vs-all

[c] 1800 is the closest option to 1793.

5.

```
Question 5
C:0.001, E_in:0.015772870662460567
C:0.01, E_in:0.014401316691811822
C:0.1, E_in:0.013029762721163077
C:1.0, E_in:0.012618296529968454
```

Figure 3 In-sample error rates for different C values

[d] Maximum C achieves the lowest Ein is the correct option as can be found from the results above.

6.

```
Question 6
C:0.001, Q:2, E_in: 0.015772870662460567, # of SVs:652
C:0.001, Q:5, E_in: 0.013989850500617199, # of SVs:339
C:0.01, Q:2, E_in: 0.014401316691811822, # of SVs:386
C:0.01, Q:5, E_in: 0.013029762721163077, # of SVs:288
C:0.1, Q:2, E_in: 0.013029762721163077, # of SVs:309
C:0.1, Q:5, E_in: 0.011383897956384583, # of SVs:258
C:1.0, Q:2, E_in: 0.012618296529968454, # of SVs:281
C:1.0, Q:5, E_in: 0.010835276368125086, # of SVs:230
```

Figure 4 In sample error rates for different C and Q values

[b] When C = 0:001, the number of support vectors is lower at Q = 5 is the correct option.

Cross Validation

7.

```
Question 7
C:0.001, Q:2, E_in: 0.015772870662460567, CV:0.9842268448052314
C:0.01, Q:2, E_in: 0.014401316691811822, CV:0.9851870642839693
C:0.1, Q:2, E_in: 0.013029762721163077, CV:0.9868331548189488
C:1.0, Q:2, E_in: 0.012618296529968454, CV:0.9875190258751904
```

Figure 5 Cross validation scores for different C values

8.

RBF Kernel

9.

```
Question 9-10
C:0.01, E_in:0.00548621588259498, E_out:0.02242152466367713
C:1, E_in:0.0019201755589082431, E_out:0.02142501245640259
C:100, E_in:0.0037031957207516114, E_out:0.018933731938216243
C:10000, E_in:0.005349060485530106, E_out:0.017937219730941704
C:1000000, E_in:0.006171992867919353, E_out:0.017937219730941704
```

Figure 6 In-sample and out-sample error rates for different C values

[d] C = 10^4 has the lowest E_{in}.

10.

[d] C = 10^4 [e] C = 10^6 has the lowest E_{in} values.